

§11.4.27 benchmarking suite — measured ns/op

Go testing.B micro-benchmarks (real measured ns/op + B/op + allocs/op), the benchmarking test-type §11.4.27 requires. Complements the HTTP-level loadtest (throughput) + stress/chaos (resilience) with per-operation cost.

```
Run: cd server && go test -bench=. -benchmem -run='^$' ./internal/  
api/ ./internal/store/
```

Captured (this host, see bench.log for the full run)

Benchmark	ns/op	allocs/op
Healthz (full router GET /healthz)	~2,489	32
GroupCreate (POST /groups)	~8,537	79
GroupList (GET /groups, 20 seeded)	~14,299	78
ClientUpdate no-deployment (auth device → 204)	~4,915	49
MemoryCreateGroup	~666	4
MemoryFindDelta (100 seeded)	~160	0
MemoryListAudit (200 seeded, limit 50)	~16,331	10

Numbers are this-host measurements (Apple Silicon), not production guarantees — they are a baseline to catch regressions, captured per §11.4.6 (measured, not asserted).